1. **Scenario:** A system checks if a user is eligible to vote based on their age.

   Write logic to ask the user for their age and determine if they are eligible to vote based on whether they are 18 or older.

   Logic:
   a). Ask the user to enter their age
   b). Check if the age is 18 or older
   c). If yes, print “Eligible to Vote”
   d). Otherwise, print “Not Eligible”

2. **Scenario:** A program processes a list of numbers and needs to find the largest value.

   Write logic to identify and return the largest number from a given list.

   Logic:
   a). Read the list of the numbers
   b). Assume the first number is the largest
   c). Iterate throughout the list, compare each number with the current largest number.
   d). If yes, print other largest value than current
   e). Otherwise print the current largest value.

3. **Scenario:** A company provides employees with a 10% bonus if their salary exceeds $50,000.

   Write logic to determine the bonus amount based on the given salary.

   Logic:
   a). Read the salary data set in the ordered dictionary
   b). Find the salary value which is above $50,000
   c). If yes, print the set of name and salary amount
   d). Otherwise print no salary value found

4. **Scenario:** A program evaluates a number to determine if it is even or odd.

   Write logic to check whether a given number is even or odd.

   Logic:
   a). Ask the user to enter the number or for loop list

b). Check the “n” number is divisible by 2 and the remainder value is 0
c). If yes, print “n is even number”
d). Otherwise print “n is odd number”

5. **Scenario:** A text-processing tool reverses a given word or sentence for formatting purposes.

   Write logic to take a word or sentence as input and produce its reversed version.

   Logic:
   a). Ask the user by input “enter the word or sentence:”
   b). check the given input type() must be strings.
   c). If yes, process the input to reverse format
   d). Otherwise print “No words or sentence found”

6. **Scenario:** A grading system determines whether a student has passed or failed based on their score.

   Write logic to check if a student has passed a subject by scoring at least 40 marks.

   Logic:
   a). Ask the student input “enter the subject marks:”
   b). Check the marks is above 40 marks
   c). If yes, print “Passed”
   d). Otherwise, print “Failed”

7. **Scenario:** A retail store offers a 20% discount if a customer’s total order exceeds $100. Write logic to calculate the final amount to be paid after applying the discount.

   Logic: a). Read the list of orders and add the order total
   b). If the total amount exceeds $100
   c). Calculate the 20% discount on value count
   d). If yes, apply the discount on value count and display the bill
   e). Otherwise, print “No discounts allowed”

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8. **Scenario:** A banking system processes withdrawal requests and ensures the user has enough balance.
   Write logic to check if a user has enough balance before allowing a withdrawal and update the remaining balance accordingly.

   Logic:

   a). Ask the input details of mobile # & OTP or pin #

   b). Check the mobile # & OTP or pin # to be matched then,

   c). if not matches deny the transaction

   d). If matched, check the account balance is equal or greater than withdrawal value.

   e). If yes, process the withdrawal amount.

   e). Else account balance is less than withdrawal value

   f). Otherwise Print “Insufficient balance”

9. **Scenario:** A calendar system verifies whether a given year is a leap year based on standard leap year rules.
   Write logic to determine whether a given year is a leap year.

   Logic:

   a). Ask the input “Enter the year”

   b). Check the year “year” is divisible by 4 & 400

   c). If the remainder is equal to 0

   d). Then print “is leap year”

   e). Otherwise print “is not leap year”

10. **Scenario:** A program filters out only even numbers from a given list.
    Write logic to extract and return only the even numbers from a list.

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Logic:

a). Read the numbers from a list

b). Check if the number is divisible by 2

c). Iterate the same throughout the list, check on each number divisible by 2 and the remainder is 0

d). If yes, print the even number list

e). Otherwise print “No even number”

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